# Sanaul Haque

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Course 66

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# Research Proposal

**Topic:**

**School Infrastructure.**

**Title:**

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# Investigating the Relationship Between Classroom Infrastructure and Educational Outcomes: A Case Study of a Rural Basic School in the Asunafo North Municipal, Ahafo Region, Ghana.

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Education

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**Country:**
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**Submission Date:**
20th September 2026.

## Introduction

School infrastructure is an important component of the physical learning environment. Conditions such as classroom space, ventilation, lighting, structural integrity and furniture can shape learners' learning experiences and are associated with educational outcomes [1], [2]. In Ghana, studies have reported relationships between the physical school environment, educational infrastructure and academic performance [3], [4]. Recent evidence from rural Ghana also indicates that inadequate educational infrastructure is associated with poorer performance outcomes among basic school learners [5]. However, much of the existing research examines infrastructure broadly, providing limited evidence on how the actual physical condition of classroom buildings relates to learners' educational outcomes within an individual rural school.

This study addresses this gap through a case study of a rural basic school in Asunafo North Municipality, Ahafo Region, Ghana. The study will assess the physical condition of the school's classroom buildings and examine its relationship with learners' educational outcomes. The study is relevant to the municipality because previous research has identified school-level factors as important to educational outcomes in Asunafo North [6], while rural schools in the municipality continue to experience infrastructure challenges. By combining direct observation of classroom conditions with learners' examination results and information from learners, teachers and the headteacher, the study seeks to provide context-specific evidence that can inform classroom maintenance, rehabilitation and educational infrastructure decisions.

## Research Questions

The study will address the following questions:

1. What is the condition of the physical school infrastructure in the selected rural basic school in Asunafo North Municipality?
2. What relationship exists between the condition of physical school infrastructure and learners' educational outcomes in the selected rural basic school?

## Literature Review

School infrastructure forms part of the physical environment within which teaching and learning occur. International evidence suggests that physical learning environments can influence learners' experiences and educational outcomes, although the strength of the relationship varies according to context and other school-related factors [1], [2]. In Ghana, Baafi found that the physical school environment was significantly related to students' academic performance [3]. Agyei *et al.* similarly reported a relationship between education infrastructure inequality and academic performance at the district level [4]. Yahaya *et al.* provide particularly relevant rural evidence, finding significant associations between educational infrastructure and performance among basic school learners in rural Ghana [5]. Nevertheless, these studies do not specifically isolate the condition of classroom buildings within one rural school and relate directly observed building conditions to learners' examination outcomes. This study addresses that gap through direct infrastructure observation combined with learner outcome data.

## Proposed Methodology

The study will employ a case study design with a quantitative component, focusing on one selected rural basic school in Asunafo North Municipality. The study will examine the physical condition of the school's classroom buildings and relate the observed conditions to learners' educational outcomes.

Physical infrastructure will be assessed using a researcher-administered observation checklist. The checklist will systematically assess observable characteristics of the classroom buildings, including the condition of walls, roof, floor, doors and windows, lighting, ventilation, classroom space, desks and furniture, maintenance, accessibility and safety. Each indicator will be rated using defined condition categories, allowing an overall description of the physical condition of the classrooms.

Data on educational outcomes will be obtained from school records on learners in the selected classes through their recent examination averages. Learners will also complete a short questionnaire capturing their experience of the classroom learning environment. Teachers will provide information about the classroom in which they teach, while the headteacher will provide school-level information on the buildings and their use. Using these complementary sources will enable triangulation between observed infrastructure conditions, learner experiences, teacher perspectives, school-level information and examination performance.

The selected school will have more than one classroom structure in dissimilar conditions. The accessible population will comprise learners in the selected upper primary classes who are considered sufficiently mature to participate meaningfully in the study. Proportionate stratified sampling will be used to select the learner participants. The eligible learners will first be divided into strata according to class. The number of participants selected from each class will be proportional to the class population. Within each class, the required participants will then be selected using simple random sampling.

Descriptive statistics, including frequencies, percentages, means and standard deviations, will be used to summarise scores from infrastructure conditions and learners' examination averages. Pearson's correlation coefficient will be used where assumptions are met, to examine the association between overall infrastructure-condition scores and learners' examination averages.

Qualitative responses from teachers and the headteacher will be analysed using reflexive thematic analysis, guided by Braun and Clarke's six-phase framework [7]. The analysis will adopt an inductive approach, beginning with familiarisation with the responses and systematic coding of relevant statements. Related codes will be grouped into candidate themes, which will be reviewed, refined and defined to identify recurring patterns concerning classroom infrastructure, teaching and learning experiences, and perceived effects on learners' educational outcomes. The qualitative findings will subsequently be used to contextualise the quantitative findings.

The study will not interpret any observed association as proof that classroom infrastructure causes differences in academic performance. Other factors including teaching, prior achievement, attendance and learner characteristics may also influence educational outcomes. The case study

4 design is therefore intended to establish the nature of the relationship within the selected school and provide context-specific evidence rather than make causal or national-level claims.

**Expected Outcome**

The study is expected to provide a systematic assessment of the physical condition of classroom buildings in a rural basic school and determine whether differences in infrastructure condition are associated with differences in learners' educational outcomes. The findings may provide practical evidence for school managers, local education authorities and other stakeholders when prioritising classroom maintenance, rehabilitation and infrastructure investment in rural schools.

**Potential Limitation**

Because the research focuses on one rural basic school, its findings will not be statistically generalisable to all schools in Ghana. In addition, differences between classes may involve factors other than infrastructure. These limitations will be addressed by triangulating infrastructure observations, learner responses, teacher and headteacher information, and examination records.

**Conclusion**

The study addresses an important educational question at the intersection of the built environment and learning. By examining the actual physical condition of classroom buildings and relating those conditions to learners' educational outcomes within a rural Ghanaian school, the research seeks to generate context-specific evidence that can contribute to more informed decisions about educational infrastructure provision and improvement.

**Project Practicalities**

The study will involve consent seeking from parents, guardians and school authorities, preparation and instrument administration, physical observation of the classroom buildings, collection of questionnaire responses and examination averages, followed by quantitative and qualitative analysis. Data collection will be conducted within the selected school after obtaining the necessary institutional and school-level permissions.

The project will be conducted over 10 months. The first 3 months will be used for ethical clearance, instrument preparation, school access, data collection and data preprocessing. The subsequent 6 months will be devoted to quantitative and qualitative data analysis, interpretation and integration of findings. The final month will be reserved for writing, revising and submitting the final research report.

**Post-Programme Plan**

The findings will be developed into a full academic research report and, subject to further refinement, considered for presentation or publication in an education, development or built-environment research outlet.

## References


1. E. Edgerton and J. McKechnie, "The relationship between student's perceptions of their school environment and academic achievement," *Frontiers in Psychology*, vol. 13, 2023, Art. no. 959259. doi: 10.3389/fpsyg.2022.959259.
2. P. Barrett, A. Treves, T. Shmis, D. Ambasz, and M. Ustinova, *The Impact of School Infrastructure on Learning: A Synthesis of the Evidence*. Washington, DC, USA: World Bank, 2019. doi: 10.1596/978-1-4648-1378-8.
3. R. K. A. Baafi, "School physical environment and student academic performance," *Advances in Physical Education*, vol. 10, no. 2, pp. 121–137, 2020. doi: 10.4236/ape.2020.102012.
4. E. A. Agyei, S. K. Annim, B. Y. S. Acquah, and J. Sebu, "Education infrastructure inequality and academic performance in Ghana," *Heliyon*, vol. 10, no. 14, 2024, Art. no. e34041. doi: 10.1016/j.heliyon.2024.e34041.
5. M. Yahaya, I. Braimah, A. Adu-Gyamfi, and M. A. Nanor, "Built environment of basic schools and performance outcomes in rural Ghana: Learners' voice," *Discover Education*, vol. 4, Art. no. 96, 2025. doi: 10.1007/s44217-025-00483-w.
6. F. Ewulley, M. A. Anlimachie, M. K. Abreh, and E. E. Mills, "Understanding the nexus of school types, school cultures and educational outcomes and its implication for policy and practice," *International Journal of Educational Research*, vol. 121, 2023, Art. no. 102237. doi: 10.1016/j.ijer.2023.102237.
7. V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, 2006, doi: 10.1191/1478088706qp063oa.